Validation & Verification

Software Quality

External quality

the stakeholder point of view

Internal quality

the development team point of view

External Quality

What the stakeholders outside of the development team perceive

- Functional (hard to specify and to check)
  - Validity w.r.t. the user requirements
- Non-functional (usually easier to check)
  - Performance: scalability, Throughput
  - Usability: easy to understand, fast to use
  - Re-utilisability: can parts be reused for other software
  - Portability: is it versatile from a platform perspective
  - Interoperability: does it integrate easily with 3rd parties
  - Robustness: Is it resilent when encountering unexpected behaviors

Internal Quality

What the development team perceives of the software

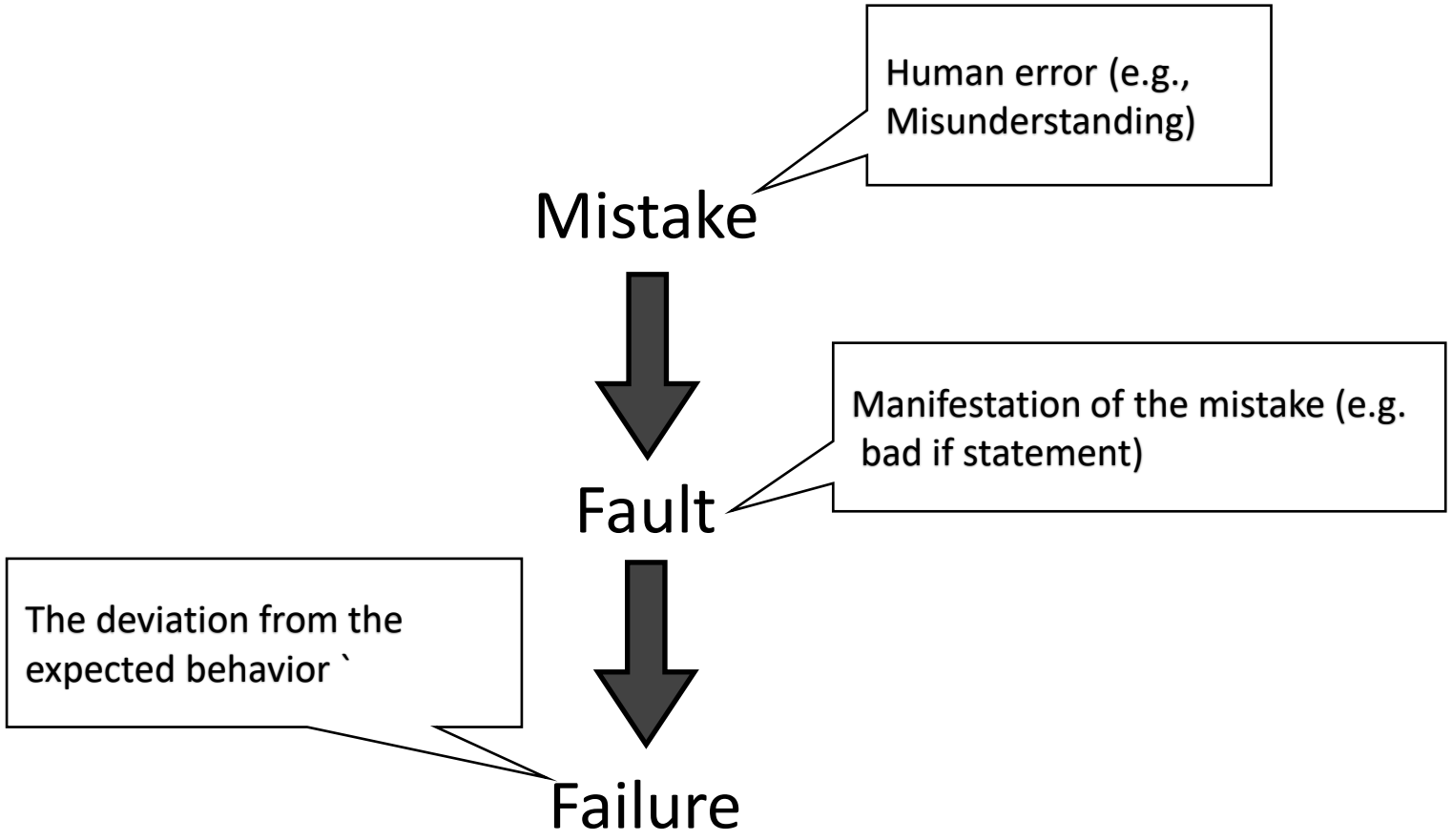
- Functional
  - Compliance with respect to the specifications
- Non-Functional
  - Verifiability: Is the software easily testable
  - Maintainability: adding a new feature or fixing a bug does not require huge preparation

How to improve/ensure quality?

Quality assurance is the result of several activities:

- Test: Exercising the software to check whether is behaves correctly w.r.t. to the specifications or the user requirements
- Model Checking: Checking that a model of the software respects specific properties
- Code Review/Audit: Staring at the code until you can read the matrix
- By Following known good practices/patterns: Standing on the shoulders of giants

By the way … what do we chase?



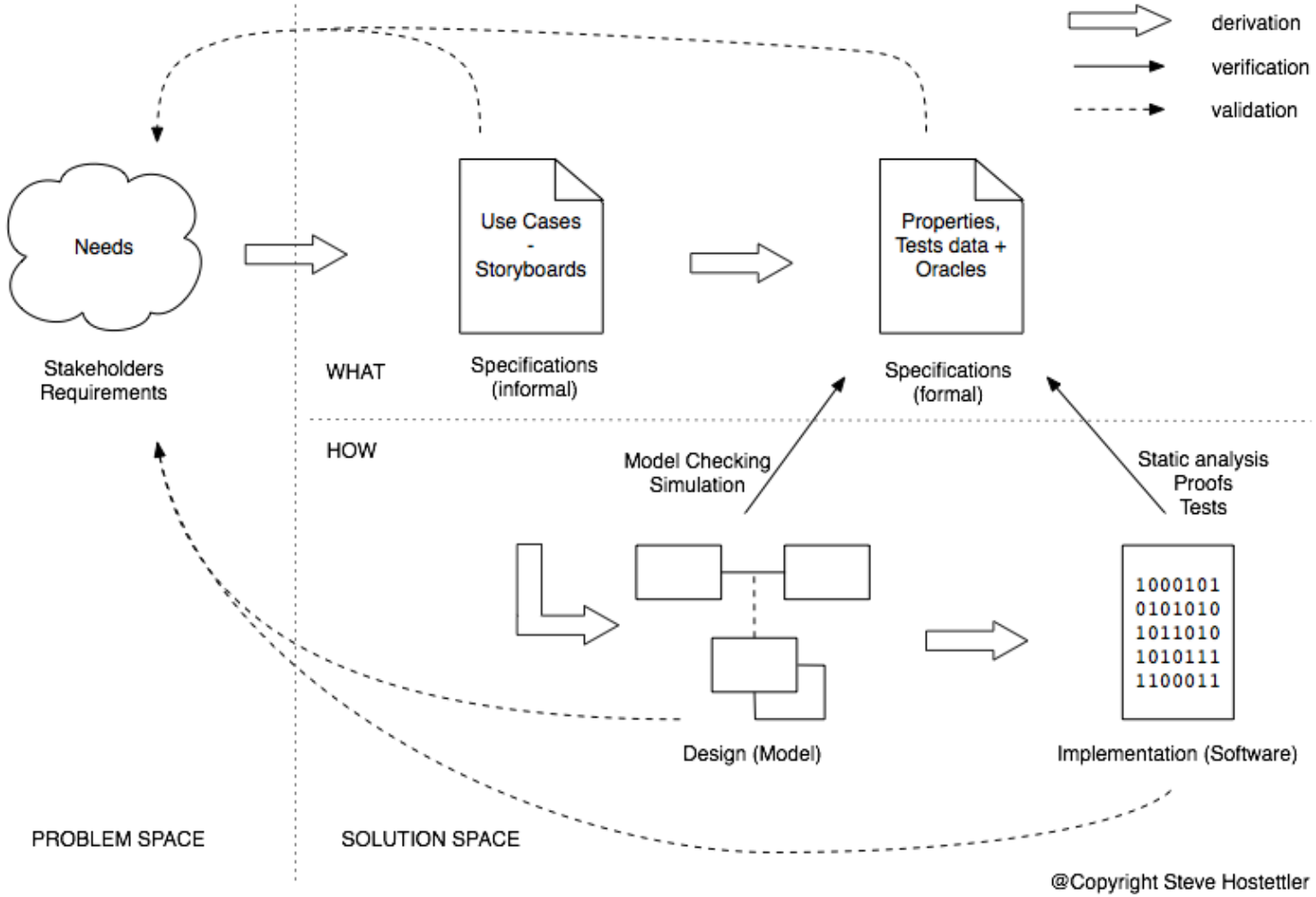
V&V

Assessing that the produced software respects the specification is not the same thing as checking that is satisfies user needs:

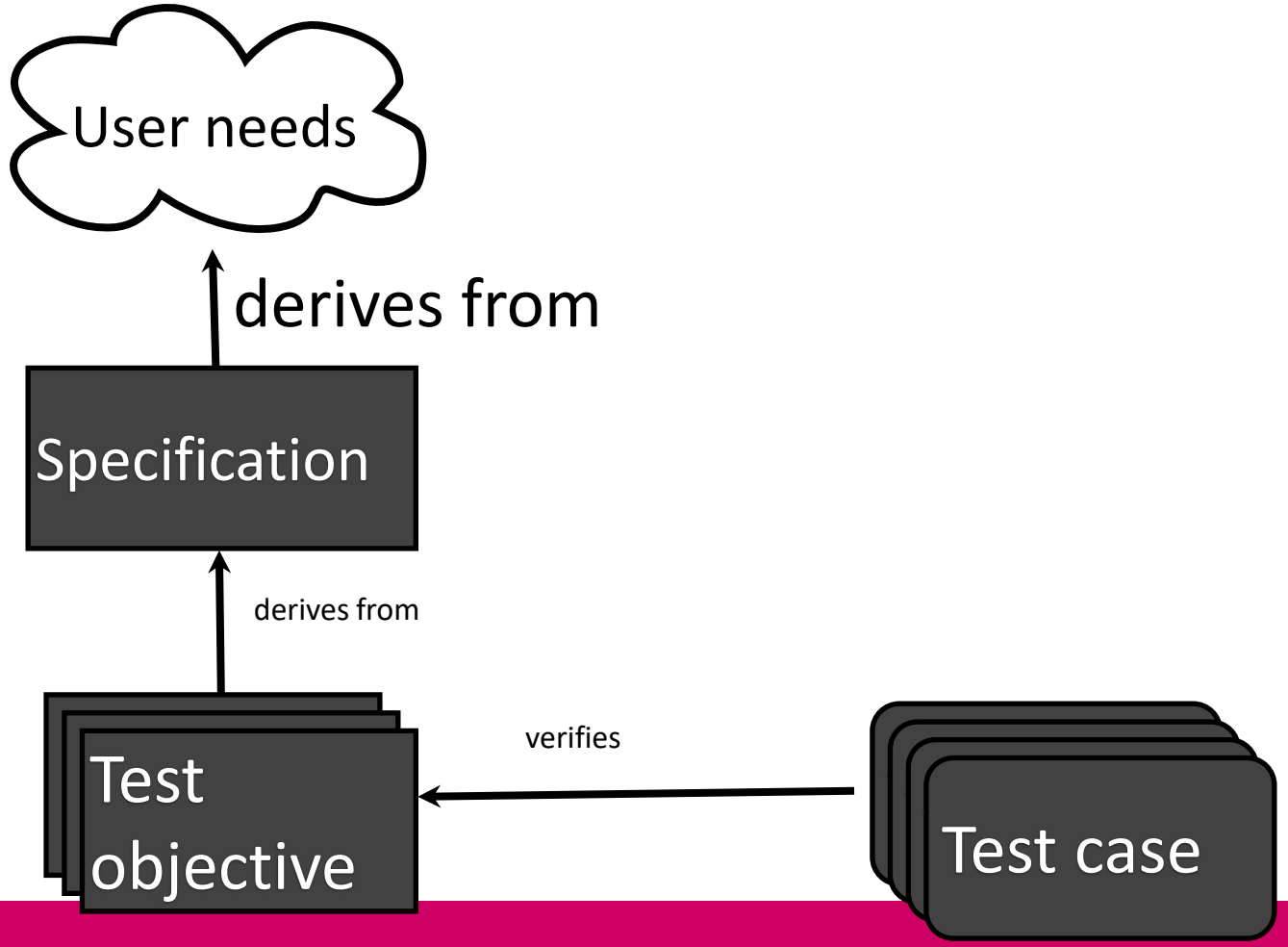
<u>Validation</u> : Did we built the right software (w.r.t. the user needs)

<u>Verification</u> : Did we built the software right (w.r.t. the specification)

Validation & Verification



Test conditions & Test cases



Testing : Vocabulary from **IEEE STD 829-2008**

- Test objective : what to test (functionality, NFR)
- Test Case : verifies a test objective and consists in a set of input values, pre-conditions, expected result and post-conditions
- Test Procedure : procedure to setup, execute, and evaluate the result of a sequence of test cases
- Test plan : description of the scope, approach, resources and schedule of the test activities.

What is testing

- Testing consists in the following activities:
  1. Tests planning and control
  2. Test analysis and design
  3. Test implementation and execution
  4. Evaluate exit criteria and reporting
  5. Test closure activities

Test planning and control

1. Definition of the test objectives
2. High-level scheduling
3. Control and reporting of deviations

Test analysis and design

- Transform general test conditions into real test cases:
    - Put the test architecture in place
    - Designing high-level test cases

Test implementation and execution

- Prioritizing Test cases implementation
- Creating test procedure
- Creating a test harness (environment)
- Check bi-directional traceability

Test implementation and execution

- Executing the test procedure
- Logging outcome
- Compare actual and expected results
- Report failure and trace back the reason
- Repeat the execution to check that the fault has been fixed.

Evaluate exit criteria & reporting

Collect the data produced by the test executions and present them to the stakeholders so that they can make an informed decision.

Evaluate the confidence in the software quality

Test closures activities

- Improve test infrastructure and guidelines based on the experience
- Clean up the test infrastructure for the next run

How much testing is enough?

- Risks
- What is at stake (human life, granny's pictures, billions of $$)
- Testing should provide enough information and confidence to the stakeholder such that they can decide whether to go live.

7 principles in tests

- Testing proves the presence of defects.
- Testing DOES NOT PROVE the absence of defects (see principle 2).

(1) from ISTQB syllabus 2011